

Jagraj Gill

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Education

Simon Fraser University – Bachelor of Science in Computing Science

Dec 2027

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, x86 Assembly

Frameworks: React, Node.js, Express.js, REST APIs, GraphQL, FastAPI, CI/CD, Agile

Databases & Tools: Git, Docker, MongoDB, MySQL, SQLite, AWS S3, Linux, Confluence, ESP-IDF

Embedded Systems: STM32, ESP32, ARM Cortex, RTOS, I2C, UART, TCP/IP

Experience

Software Engineer Intern, Savi Finance – Toronto, ON

May 2025 – Dec 2025

- Built the core AI logic and MongoDB schema for a task-completion feature that interpreted prioritized to-do items and routed them toward fulfillment through third-party APIs
- Built feature parity across React and React Native codebases by extending shared GraphQL APIs with mobile-specific schema fields for 1000+ users
- Reduced image load failures by 10% by building an image upload feature that sent files to AWS S3 and persisted returned image references via GraphQL for reliable preview rendering
- Strengthened codebase integrity by writing GitHub Actions checks that flagged unauthorized package and config changes before merge

Projects

AI-Powered Study Assistant – Python, JavaScript, FastAPI, Gemini API

github.com/jagrajg/AI-Study-pal

- Built a pipeline using PyPDF and a structured LLM prompt to convert raw PDF text into flashcard datasets, automating study material generation from unstructured documents
- Designed a strict JSON schema requiring the AI to return all flashcard fields, reducing incomplete output
- Built Python validation to catch empty fields and nonsensical content, filtering bad data before it reached the user-facing flashcards
- Improved study workflow by developing an interactive JavaScript flashcard interface with flip animations and navigation controls, allowing users to review AI-generated cards in real time

Smart Environmental Monitoring System – C, JavaScript, SQLite, FastAPI

github.com/jagrajg/ESP32-IoT

- Built a full-stack IoT monitoring system that streamed live temperature and humidity readings from an embedded sensor device to a web dashboard, combining a REST API backend with persistent storage
- Developed a FastAPI backend that ingested structured JSON sensor data over Wi-Fi, storing readings in SQLite and powering a JavaScript dashboard with live readings and 24-hour trend views
- Designed fault-tolerant data ingestion that handled dropped connections and failed transmissions without data loss, surfacing real-time system status instead of failing silently

Portfolio Performance Tracker – Java, Swing

github.com/jagrajg/InvestmentTracker

- Built a Java Swing application where users could log stock holdings to visualize portfolio growth trends
- Calculated gain/loss for each stock holding, rendering results as graphs to compare performance across assets
- Added input validation to reject non-numeric or invalid entries, preventing corrupted portfolio data from broken or unrealistic inputs

Wireless Robotic Arm – C, ESP32, Embedded Systems

github.com/jagrajg/ESP-32-Arm

- Built a real-time distributed control system for a 4-DOF robotic arm, streaming sensor input from one ESP32 device to a second over a peer-to-peer wireless link to drive servo motion
- Designed a data pipeline that mapped raw analog sensor readings to per-joint servo angles, applying exponential smoothing and range constraints to prevent erratic or unsafe motion